

$H_t \rightarrow E_t$

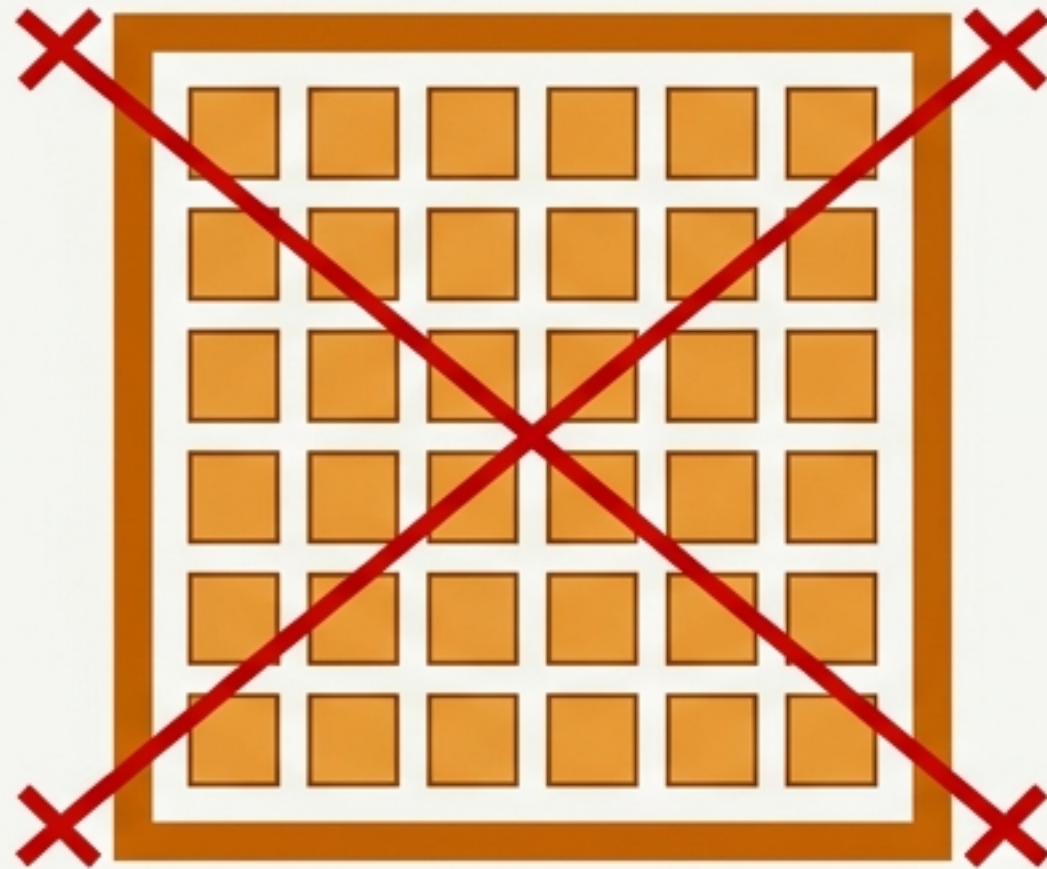
Histories Become Operators

Biological Notes on Retained Admissibility

Distilling the Levin-Schleisman Dialogue on Memory, Agency, and Learning

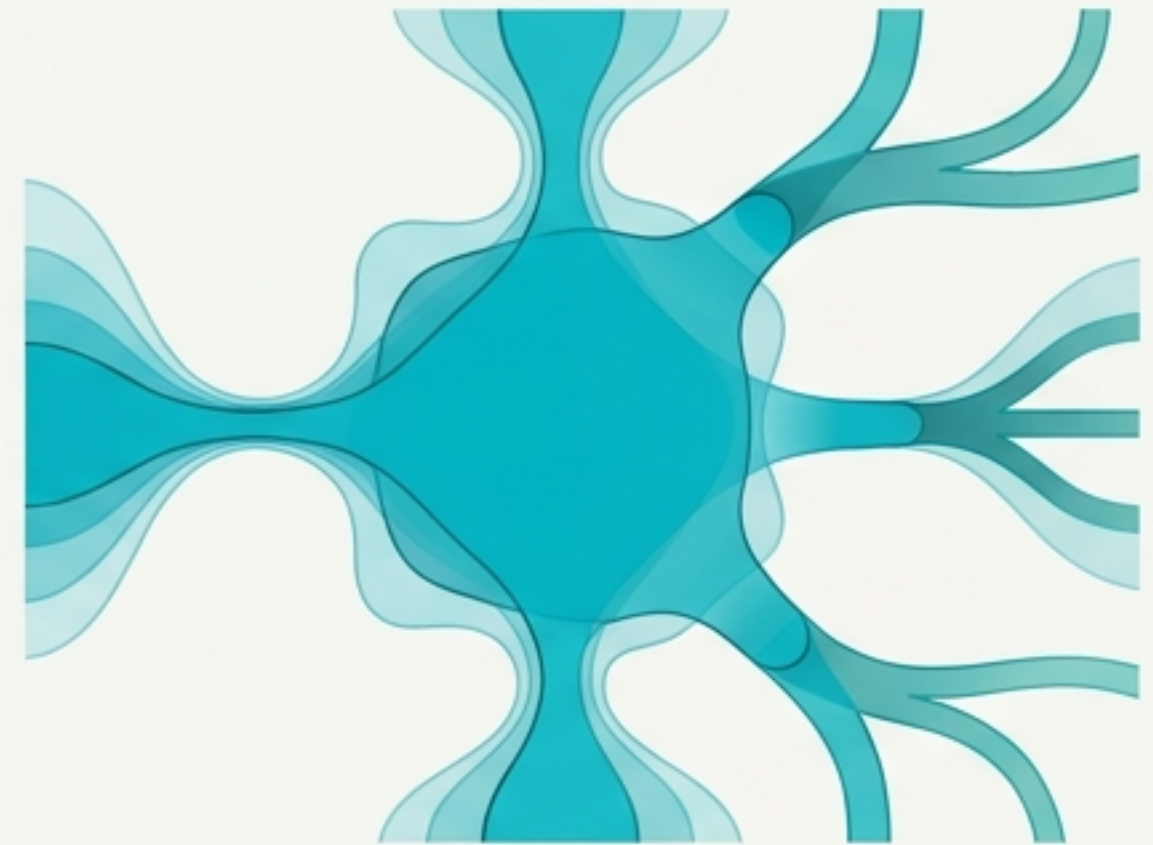
Biological memory is not a database

The Artificial Model



Engineering treats memory as a **fidelity problem**—protecting identical bits from drift.

The Biological Model

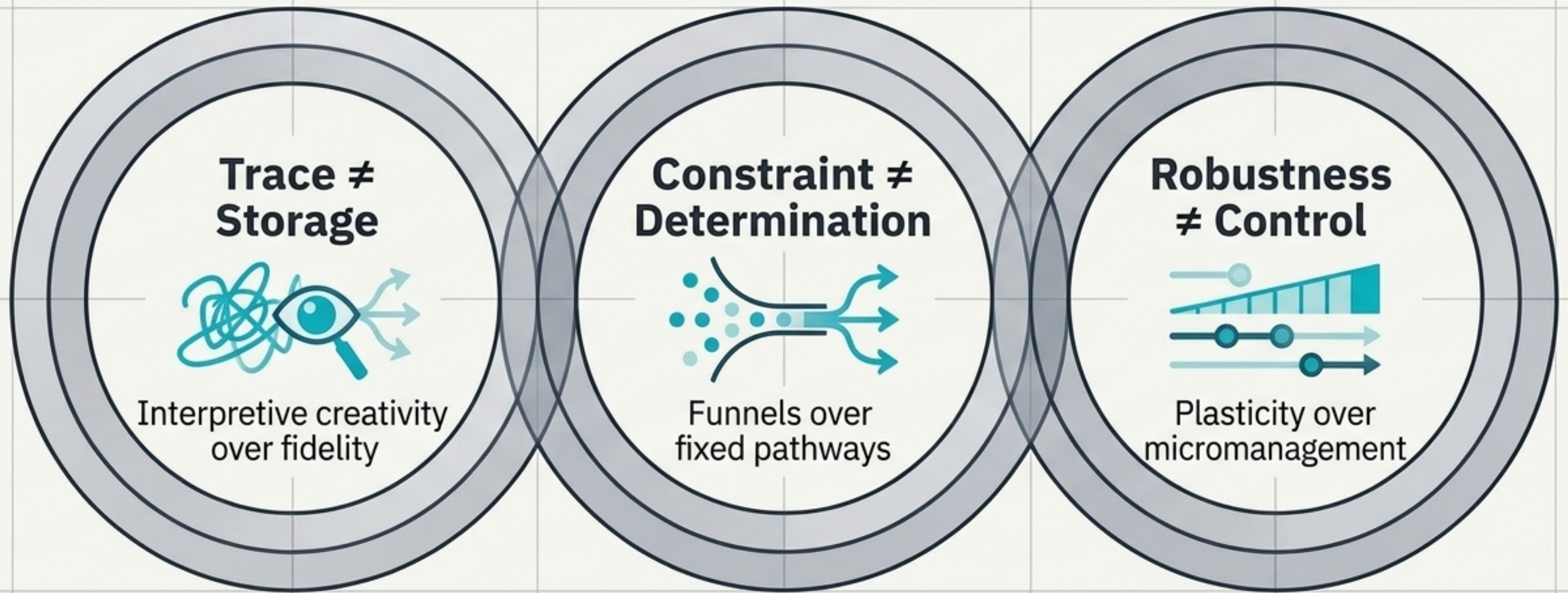


Biology treats memory as a **message** from a past self that must be actively interpreted.

Organisms differ from current AI models fundamentally: their **governing operators remain revisable** within the episode of continuation itself.

Three paradigm shifts in understanding biological systems

To integrate biological and artificial memory into a single mathematical hierarchy, we must first unlearn the engineering habits of memory design.

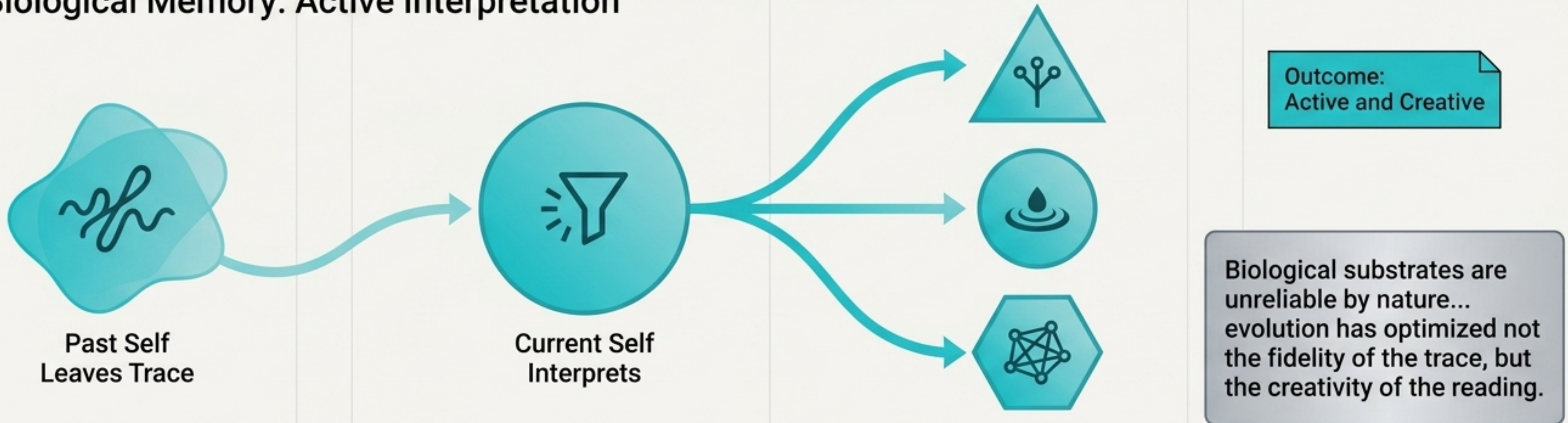


Meaning lives in the reading, not the record

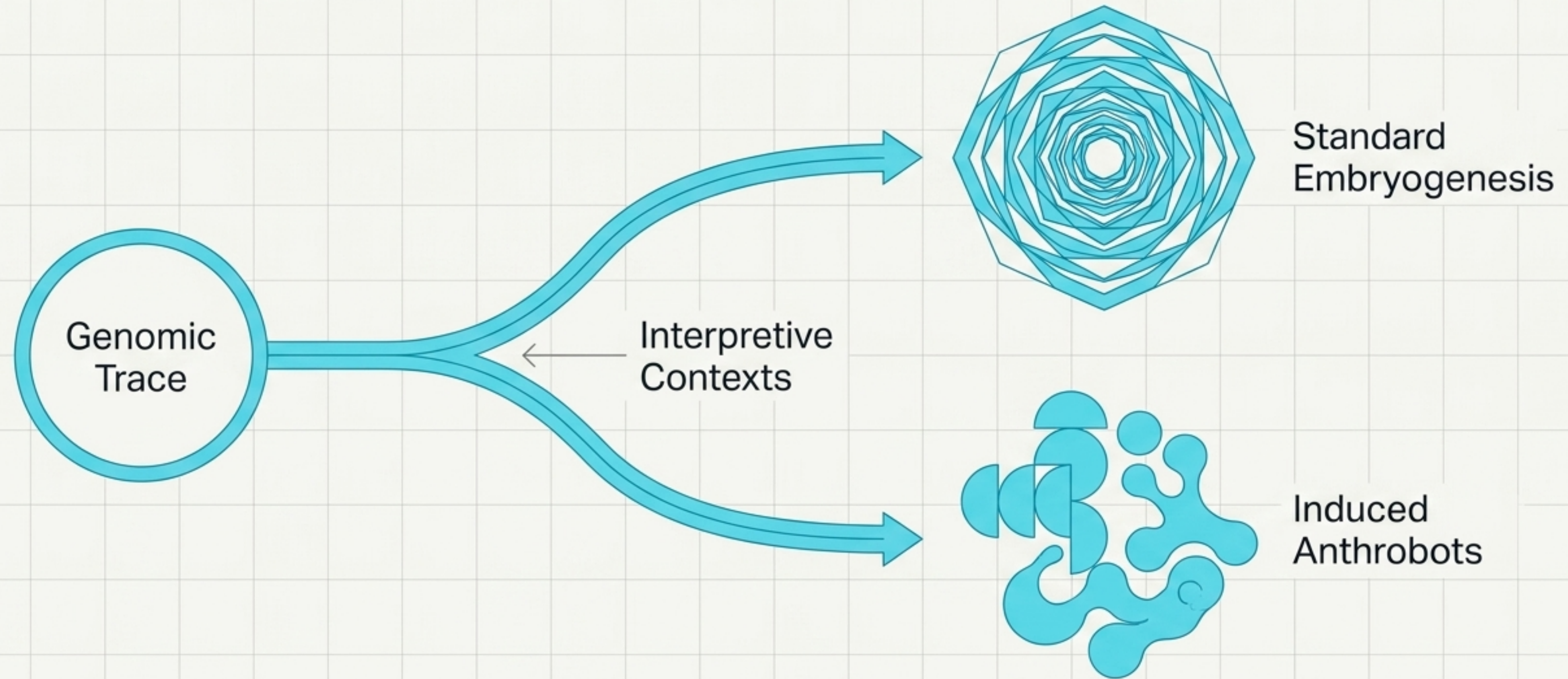
Engineering: Rigid, Amber/Burnt Orange



Biological Memory: Active Interpretation

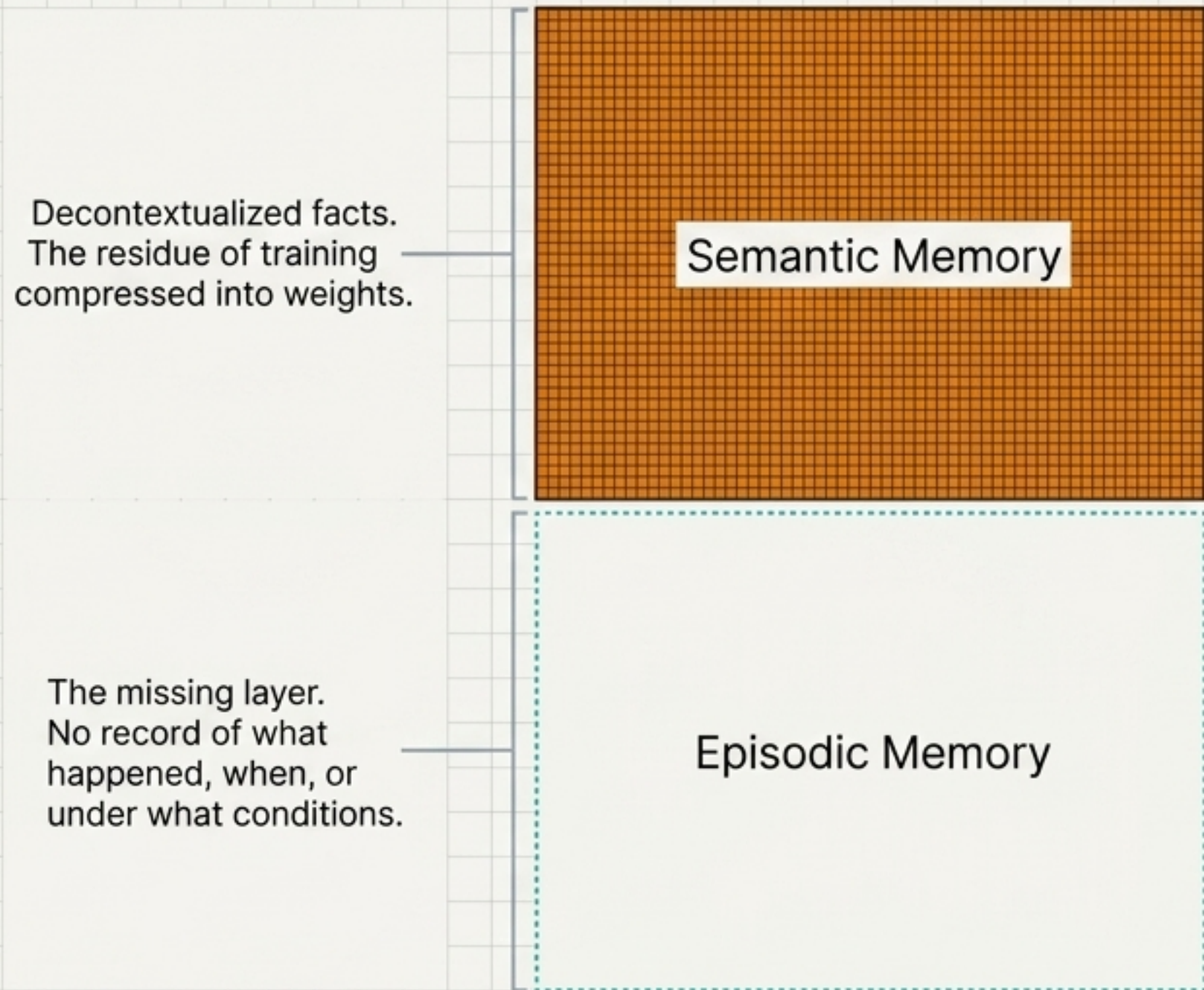


The genome is an underdetermined score, not a blueprint



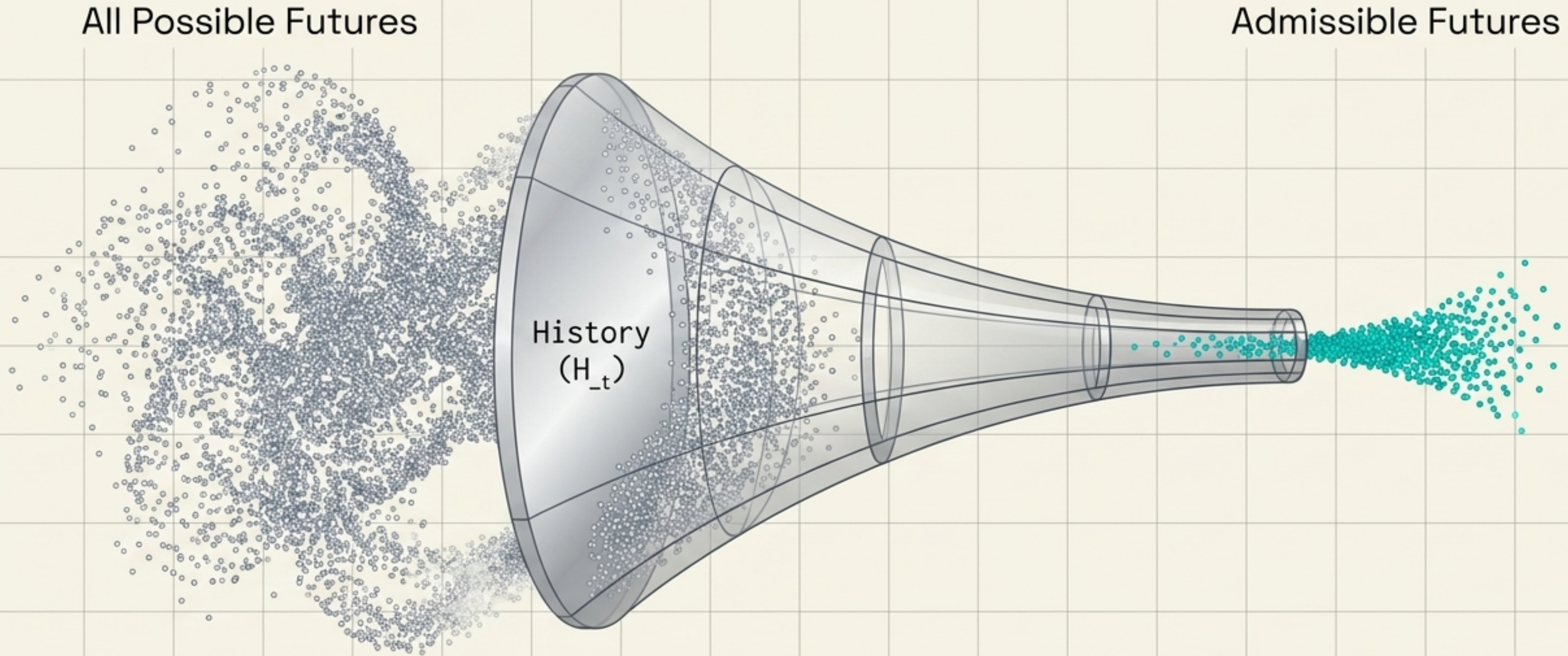
Both outcomes begin from materially identical genetic information. Thousands of genes express differently. The trace did not change; the interpretive machinery reading it did.

The missing indexicality in artificial systems



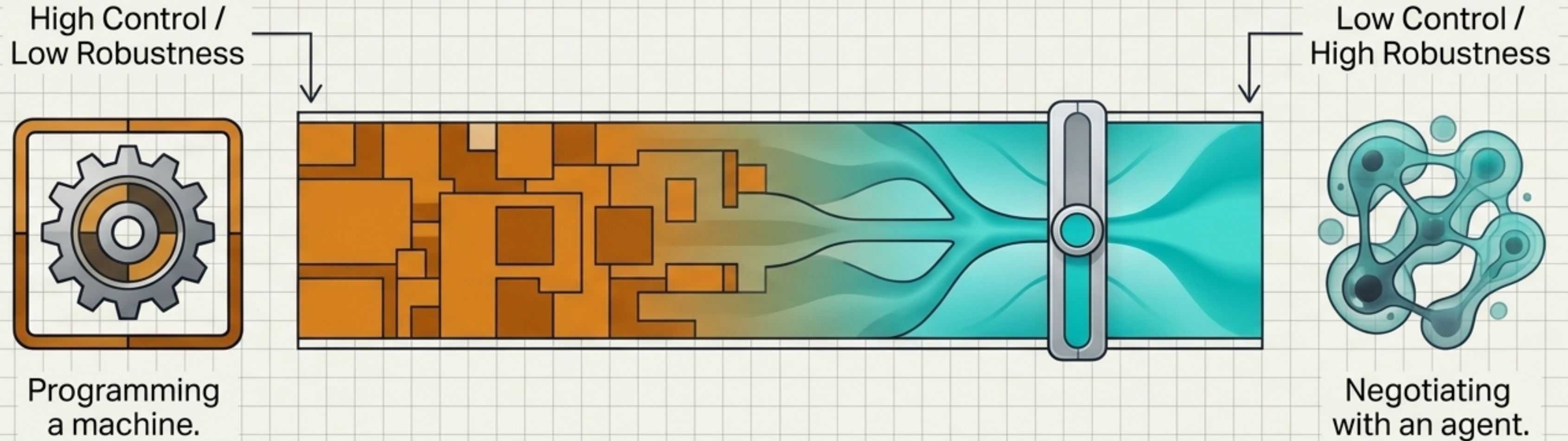
Appending conversational history via Retrieval-Augmented Generation (RAG) treats a structural symptom. It attempts to add fidelity where the true missing ingredient is indexicality (memory tied to a lived occasion).

History narrows the future without dictating it



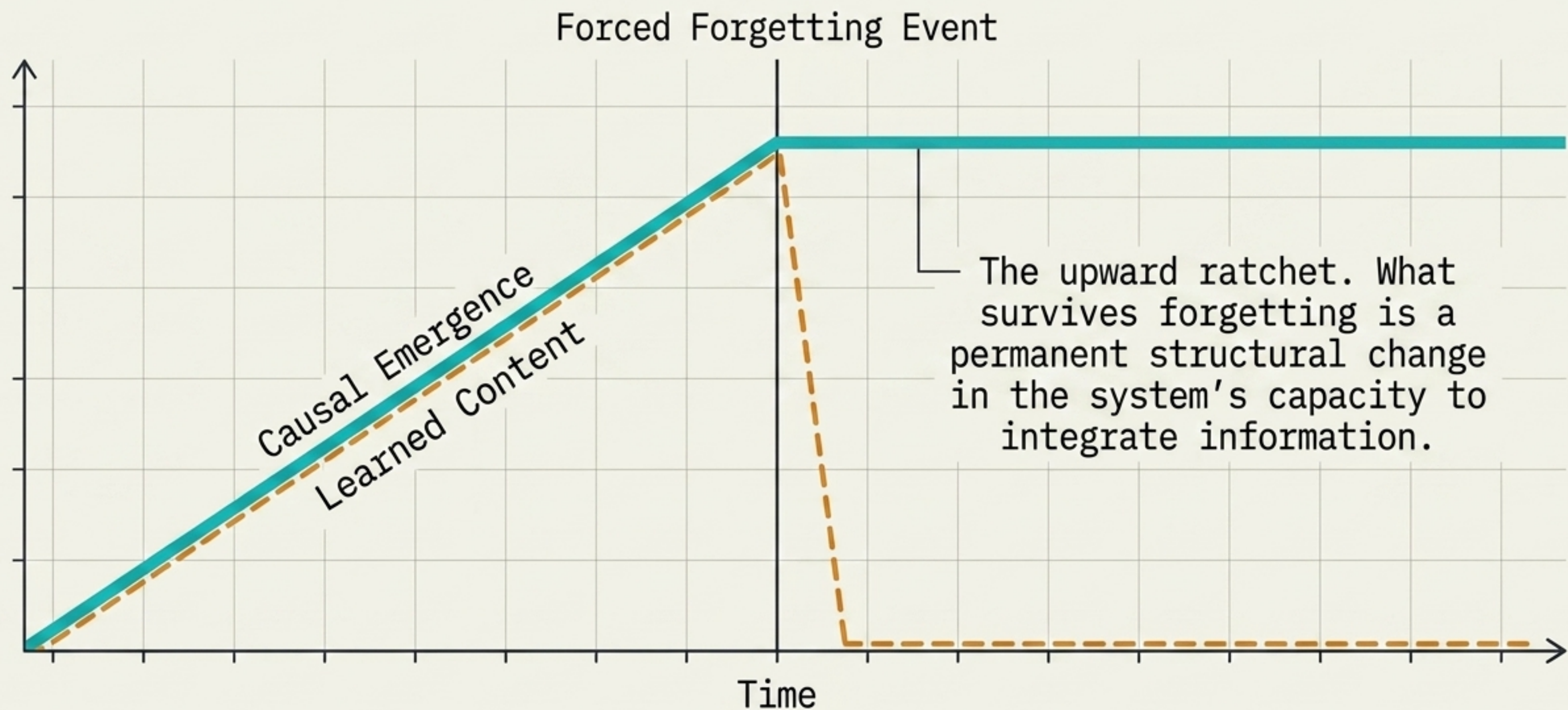
History acts as a constraint, not a lookup table. The system does not mechanically pass through this filter; the continuing agent authors the final choice in the act of reading the constraint.

Interpretation trades micromanagement for plasticity

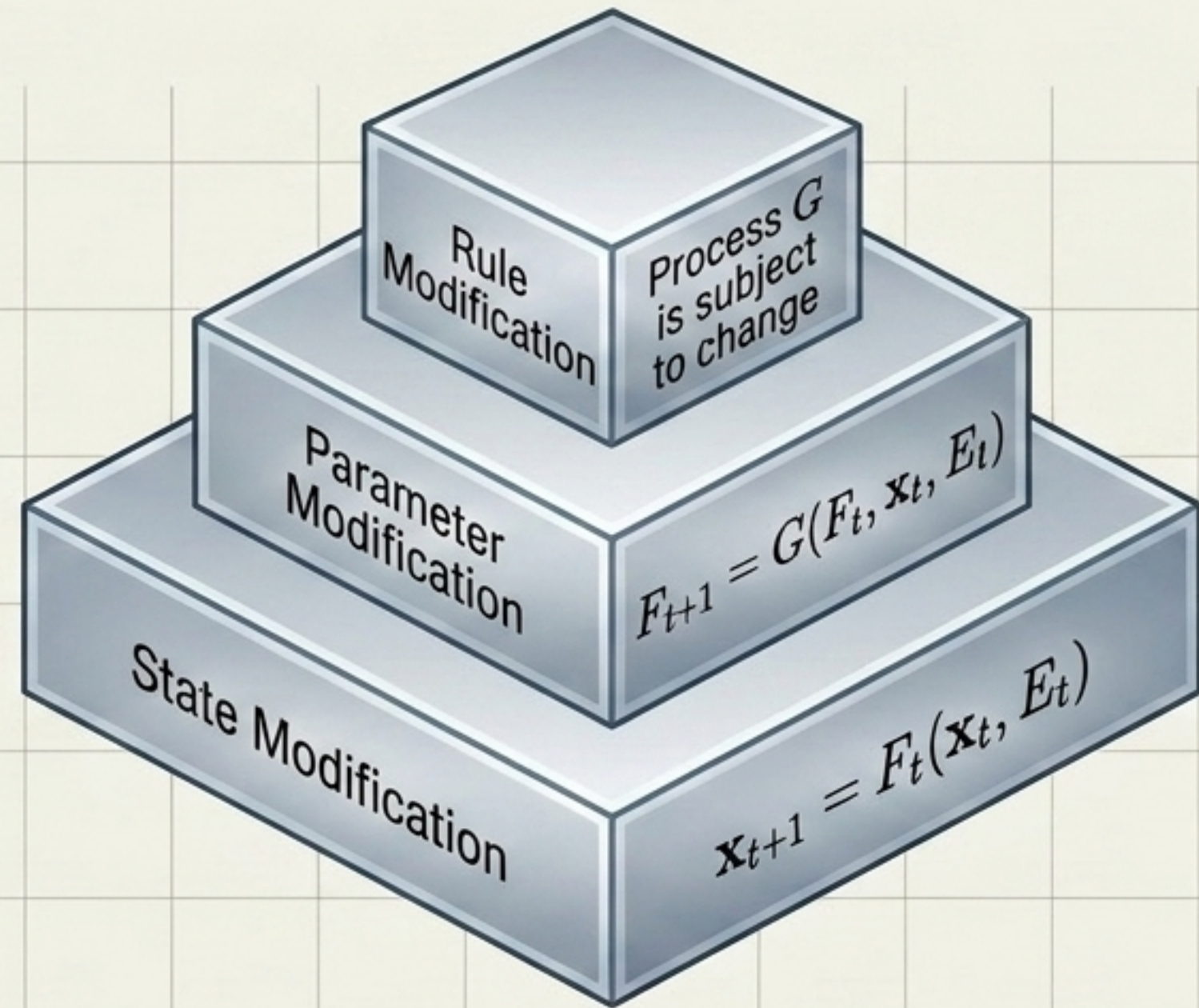


Because the organism interprets its own traces without deferring to an external reader, its behavior cannot be fully specified in advance. It is guaranteed to do something coherent with its history, but not necessarily what an external designer intended.

Structural gains survive the erasure of content



The formal architecture of recursive continuation



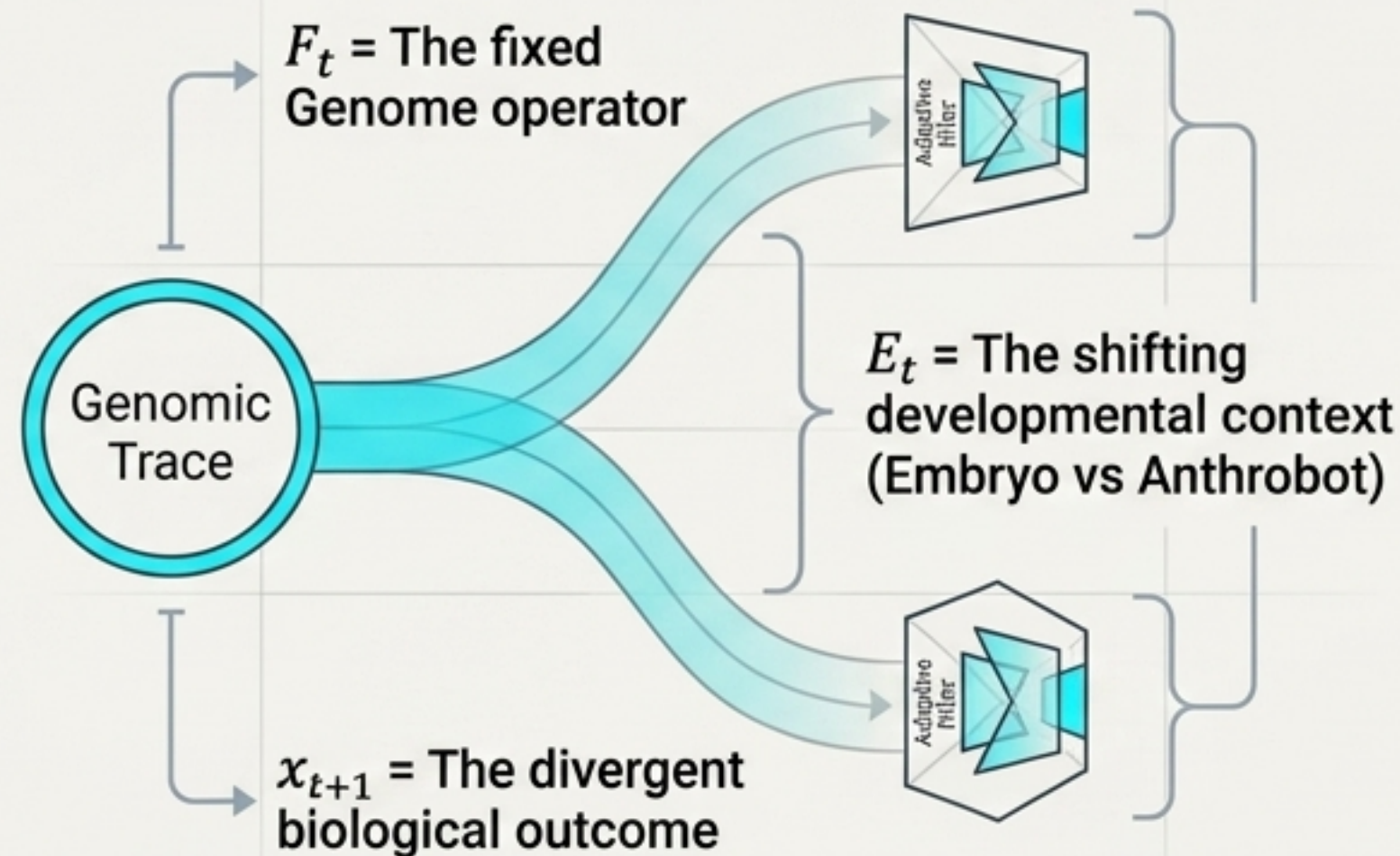
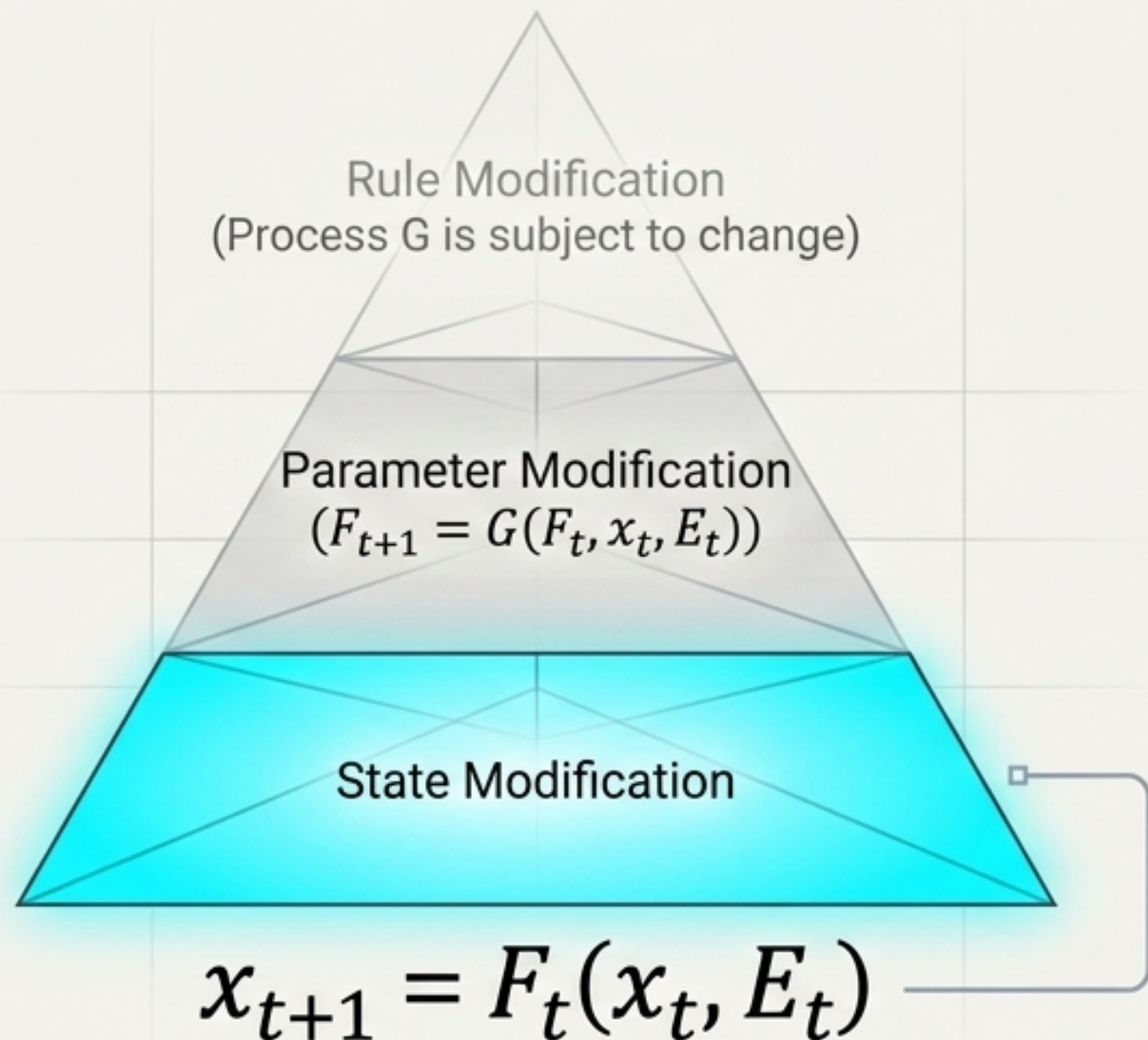
The process G that updates F_t is itself subject to change over time.

The mechanism itself is updated in light of state and environment.

A fixed adaptation mechanism acts on a changing state.

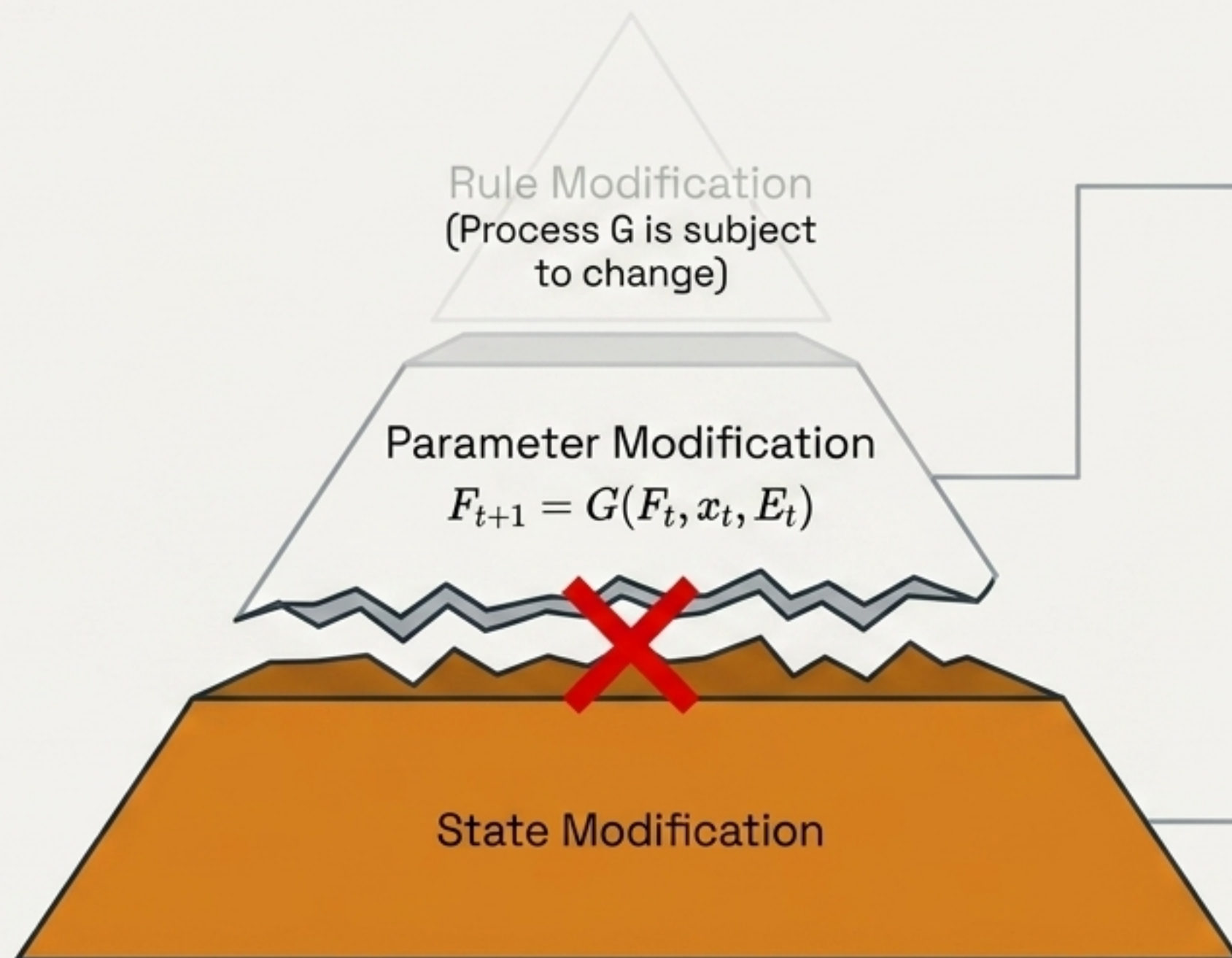
All biological instances occupy these depths of retained history.

Development modifies state, not rules



The genome case shows fixed-operator variability. The operator does not change; the context changes. History is retained only at the state level.

Language models stall at state modification

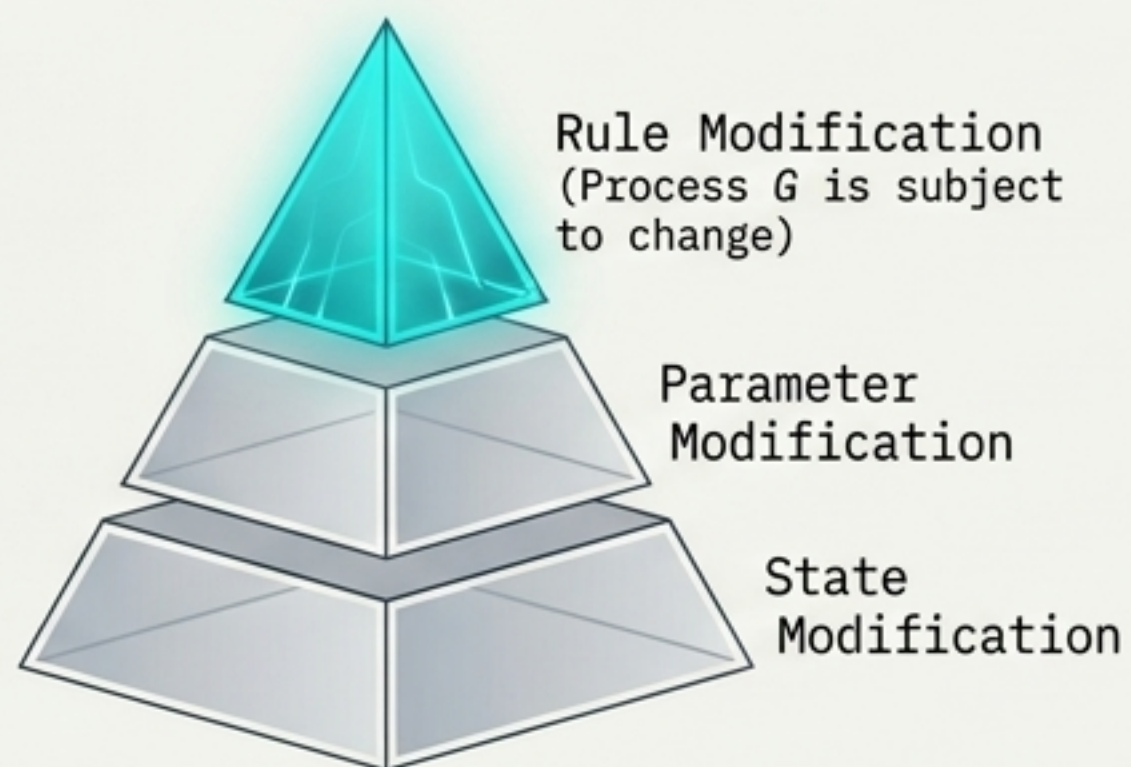


$$F_{t+1} = \cancel{G}(F_t, x_t, E_t)$$

The episodic gap in AI is the absence of an endogenous G coupled to the lived episode. Operator revision (F_t) only occurs during training (between episodes). At inference, it is an active state with no rule-modification machinery.

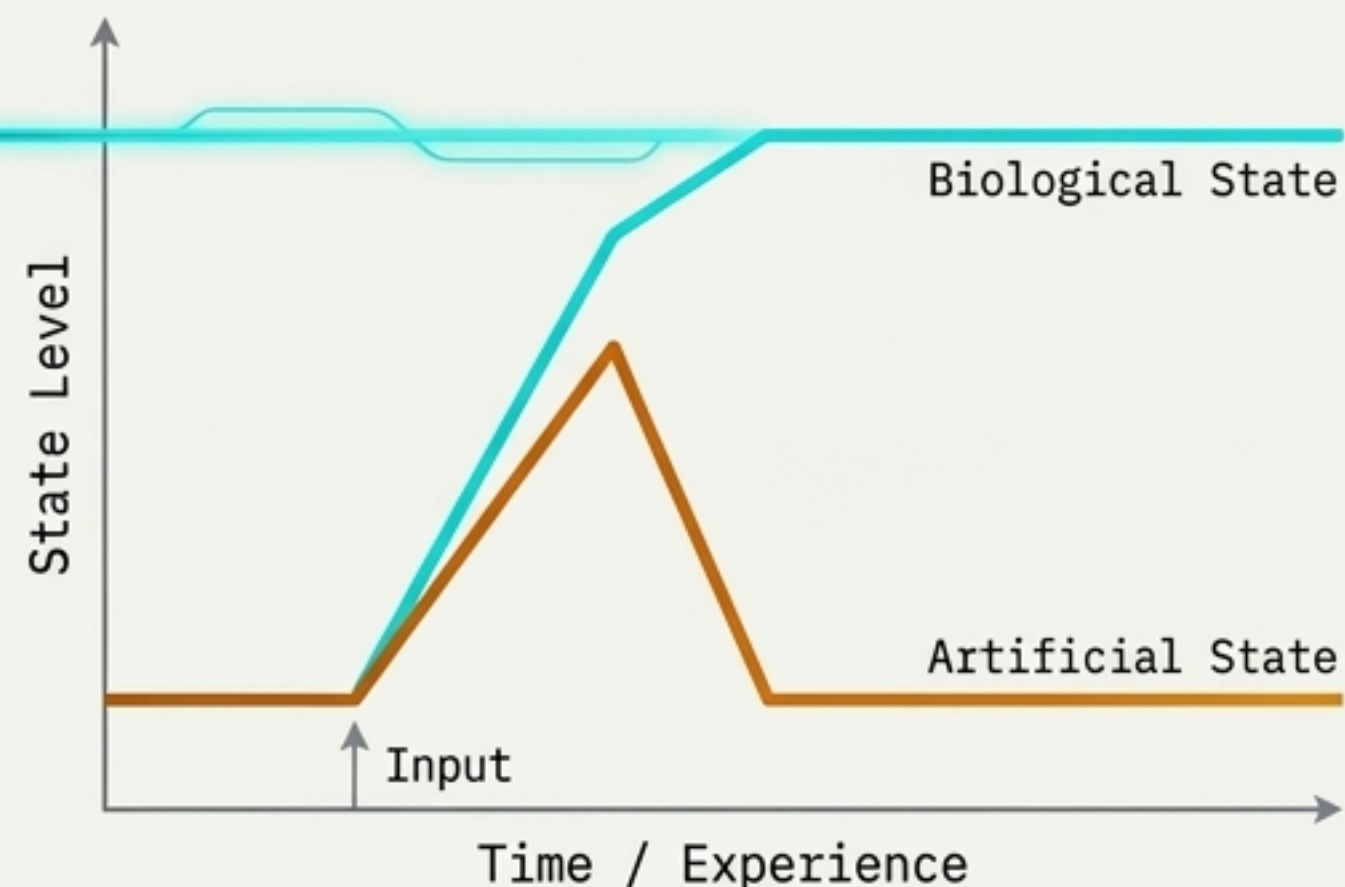
The causal ratchet permanently reshapes the operator

The Math Architecture



$$F_{t+1} = G(F_t, x_t, E_t)$$

The Biological Mapping



The nervous system operates here. The Pegozzi causal emergence result is a quantitative claim about G : the revision to F_t produced by learning is path-asymmetric. It does not symmetrically reverse when the state-level content (x_t) is erased.

Structural divergences: Biology vs. AI

	Biological Organisms	Large Language Models
Trace Processing	• Interpretive and indexical. Meaning is actively assigned by the continuer.	• Retrieval-based semantic memory. Protected from drift, but devoid of lived context.
Admissibility	• Active constraint authored in the moment of continuation.	• Passive readout strictly dictated by heavily compressed training weights.
Hierarchy Depth	• Operates fluidly through State, Parameter, and Rule modification depths.	• Frozen at State modification during inference (lacks an endogenous G mechanism).

Under what conditions does G
inherit irreversibility from H_t ?

Is the asymmetric ratchet a general mathematical property of parameter modification, or a unique condition of biological networks? The transition from function back to history remains the open edge of the admissibility program.